Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-I Biology Examination April/ May 2019

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part I (Class IX) Biology. It contains comments on candidates' responses to the 2019 SSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Candidates who did not score well mostly failed to understand the demand of the question, often misinterpreting the command word and the stimulus. Furthermore, interpretation of graphs and scientific reasoning were also weak. Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Solving biological problems following the scientific method
- Interpretation of factors affecting the enzymes' activity
- Structural-functional relationship between the parts of alimentary canal
- The concept of transpiration pull

However, candidates outperformed in some concepts, such as, comparison between bacterial and plant cell and factors affecting rate of photosynthesis.

Detailed Comments:

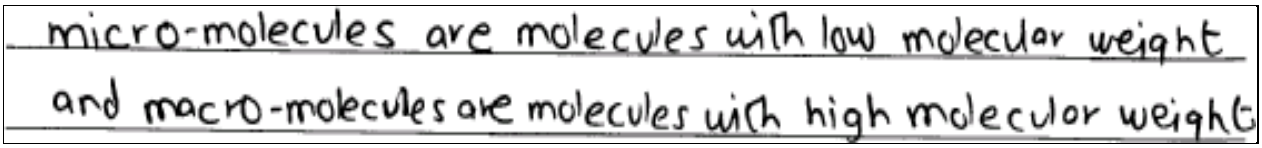
Constructed Response Questions (CRQs)

Question 1a:

Differentiate between micro-molecules and macro-molecules.

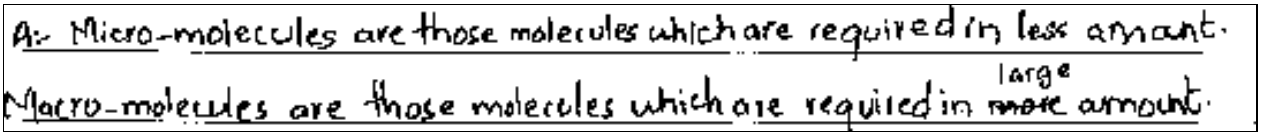
Better responses differentiated between micro and macromolecules on the basis of their molecular masses. Moreover, some candidates differentiated on the basis of their number of atoms and diffusion rate. This indicates the in-depth study of this topic.

Example:



Weaker responses were unable to differentiate between micro and macromolecules. Most of these responses differentiated on the basis of their requirement in human body. Such as micromolecules are required in less amount and macromolecules are required in large amount. However, few responses differentiated on the basis of molecular weight but swapped the meaning of micro and macro. This indicates candidates' spontaneous and careless drafting of answer.

Example:



Question 1b:

Classify the given molecules as micro-molecules or macro-molecules.

i. Starch
ii. Water
iii. Oxygen
iv. Haemoglobin

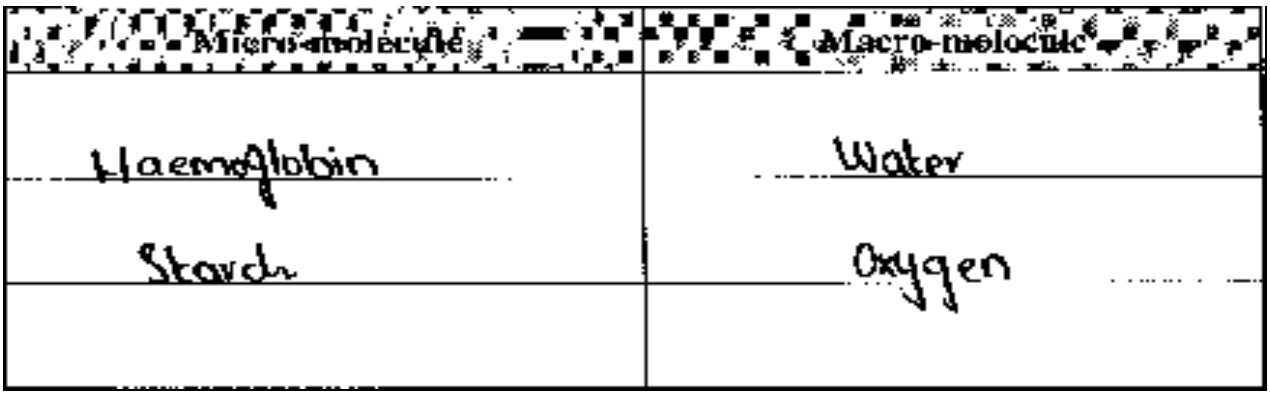
Better responses correctly classified the given examples of micro and macromolecules. For example, water and oxygen were classified as micromolecules, whereas, starch and haemoglobin as macromolecules.

Example:

Micro-molecule	Macro-molecule
Water	Haemoglobin
Oxygen	Starch

Weaker responses showed irrelevant classification of the given examples of micro and macromolecules. It was evident that those responses which failed to differentiate between micro and macromolecules were also not able to correctly classify the given examples. Candidates are advised to emphasise on the examples for better understanding of the differences.

Example:

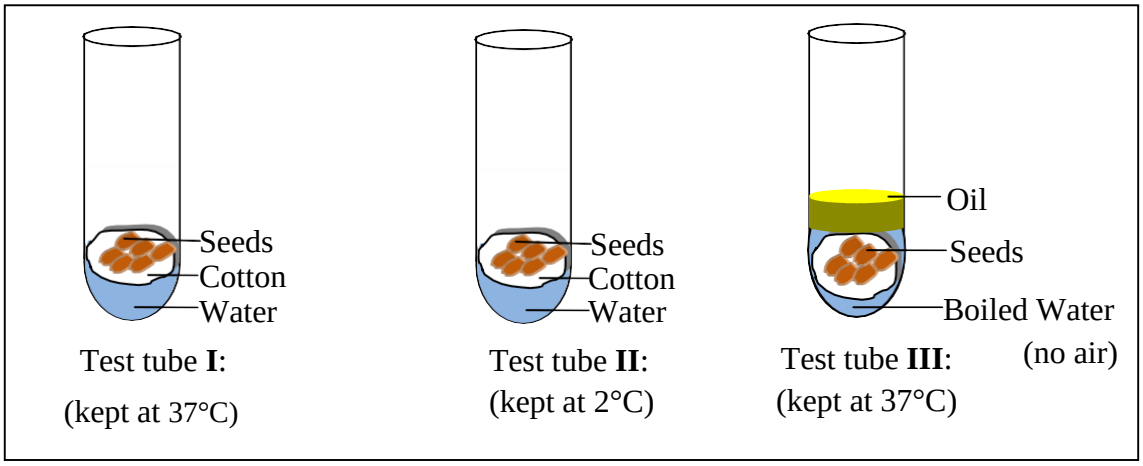


Question 2:

The given diagrams show an experiment to study the conditions necessary for germination of seeds.

Expected Result:

Seeds in test tube **I** would germinate while there would be no germination in test tubes **II** and **III**.



Explain why

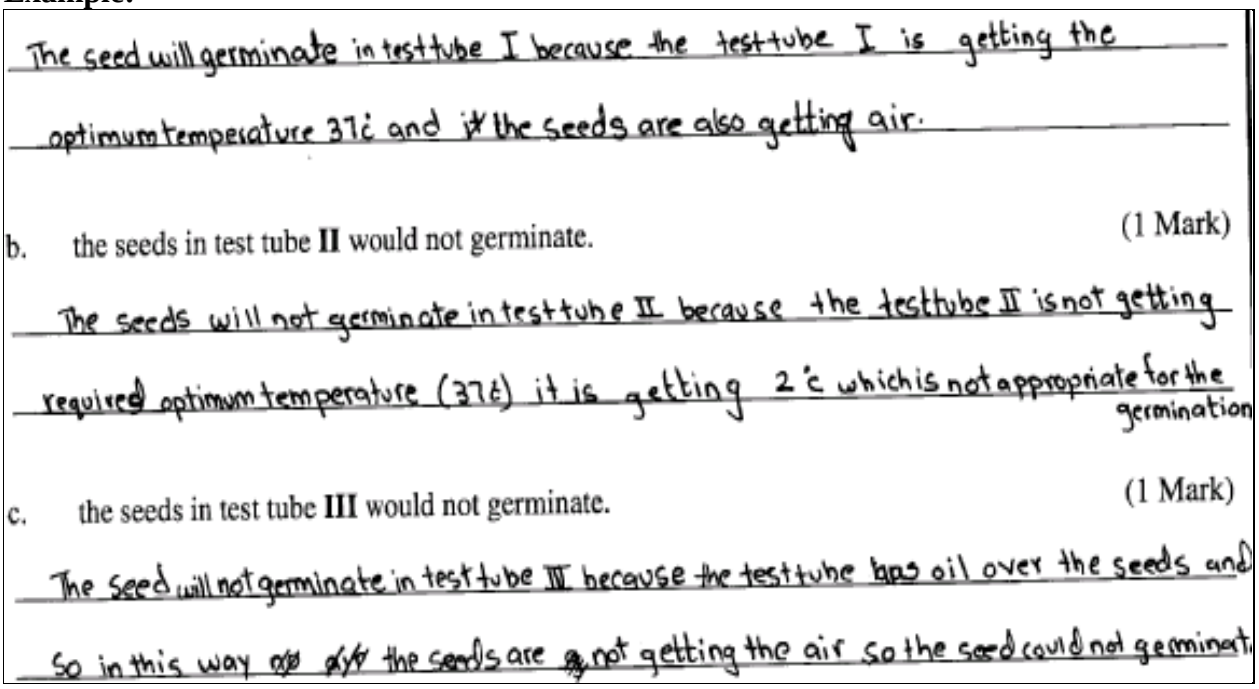
a. the seeds in test tube **I** would germinate

b. the seeds in test tube **II** would not germinate.

c. the seeds in test tube **III** would not germinate.

Better responses exhibited clear understanding of the scientific method by correctly inferring each condition and function of all given variables in the process of germination. Moreover, such responses used the scientific vocabulary, for example, (a) availability of optimum temperature and moisture in test tube I enabled the seeds to germinate, (b) unavailability of optimum temperature in test tube II caused hindrance in the germination of seeds in test tube II, and (c) in test tube III, seeds will not germinate because oxygen is absent in boiled water and the oily layer prevents further addition of oxygen into it.

Example:


The seed will germinate in test tube I because the test tube I is getting the optimum temperature 37c and the seeds are also getting air.

b. the seeds in test tube **II** would not germinate. (1 Mark)

The seeds will not germinate in test tube II because the test tube II is not getting required optimum temperature (37c) it is getting 2 c which is not appropriate for the germination

c. the seeds in test tube **III** would not germinate. (1 Mark)

The seed will not germinate in test tube III because the test tube has oil over the seeds and so in this way the seeds are not getting the air so the seed could not germinat

Weaker responses displayed a shallow understanding of the concept of scientific method. Such responses were unable to relate the stimulus with questions, thus provided vague and generalised answers. Such responses mentioned the presence of water and temperature in test tube I, absence of temperature in test tube II and presence of layer of oil in test tube III. Moreover, few responses exhibited carelessness in reading questions, thus, could not meet the demand of the question and restated the variables given in the stimulus in their answers.

However, few other responses have shown candidates' knowledge and understanding about the variables used in the question but were unable to apply their knowledge and understanding to infer the conditions. Therefore, teachers are highly recommended to use multiple examples for more practice.

Example:

a. the seeds in test tube **I** would germinate. (1 Mark)

It will germinate because it is kept at 37°C with water and is gaining heat and contains water.

b. the seeds in test tube **II** would not germinate. (1 Mark)

It would not germinate because its temperature is very than ~~it should be~~ its temperature should be.

c. the seeds in test tube **III** would not germinate. (1 Mark)

It will not germinate because it contain oil.

Question 3:

List any THREE human activities that cause deforestation.

Better responses listed the relevant human activities that cause deforestation such as urban development, industrial development, making wooden furniture, paper making at industrial scale, agricultural expansion, livestock ranching and fuel wood harvesting.

Example 1:

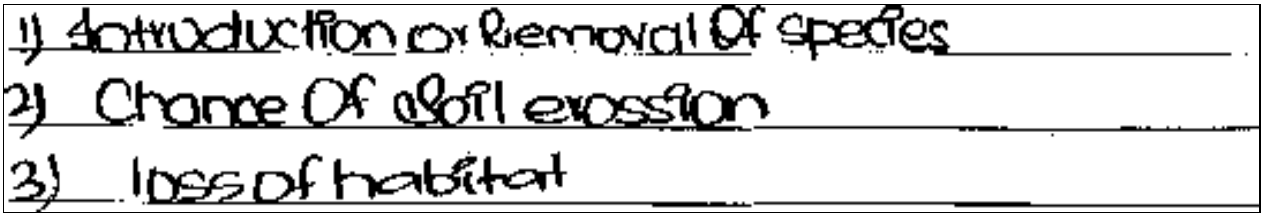
>> The large scale production of papers.
>> furniture production
>> cutting of wood for fuel purpose.

Example 2:

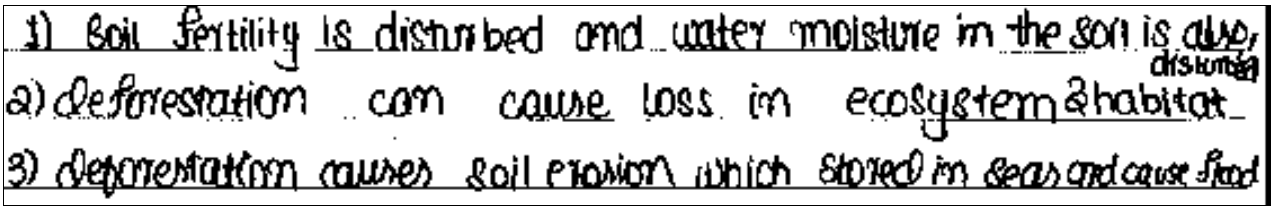
1. Cutting trees to make furniture.
2- Cutting trees to make space for construction.
3. Cutting of trees to use as firewood.

Weaker responses defined deforestation and provided answers related to the consequences of deforestation like global warming, disturbance in ecosystem and extinction of species. Such responses reveal candidates' carelessness in reading and understanding the question. Teachers are advised to inculcate in students the habit of comprehending the question.

Example 1:



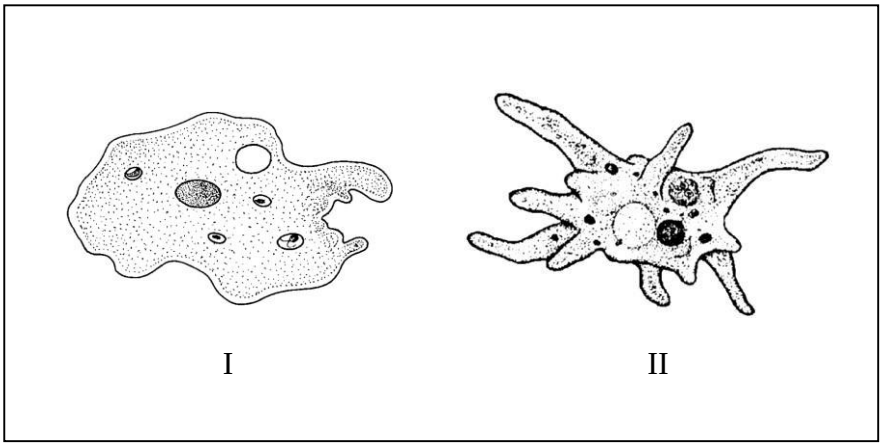
Example 2:



Better responses showed clear understanding of the concept of surface area. These responses were able to correctly identify the form of amoeba II that has greater surface area with proper description. Such responses described the features which provide large surface area, for example, presence of multiple numbers of pseudopodia/ cytoplasmic extensions/ fingerlike projections which facilitates more surface area exposed to the environment. Candidates have wisely used the diagram to draft the correct answer.

Question 4:

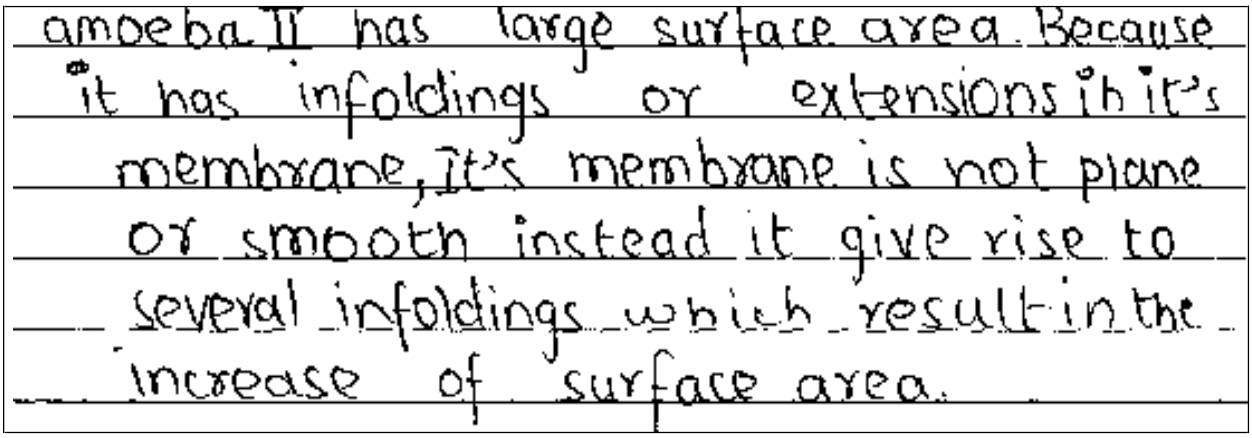
The given diagrams show two forms of the same amoebae



I II

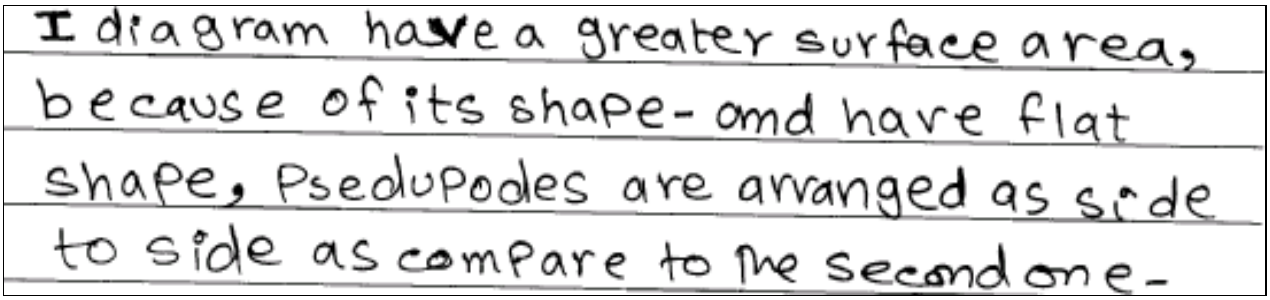
Which of the two amoebae forms, I or II, will have greater surface area? Support your answer with suitable description.

Example:

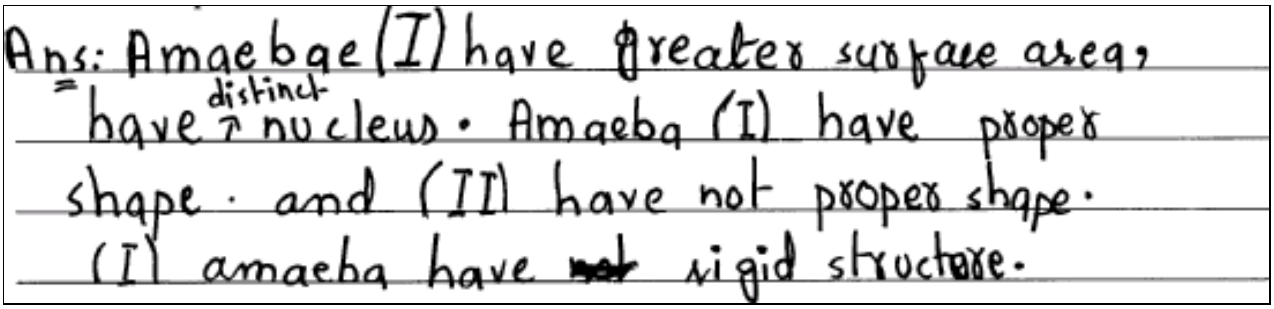

amoeba II has large surface area. Because it has infoldings or extensions in it's membrane, it's membrane is not plane or smooth instead it give rise to several infoldings which result in the increase of surface area.

Weaker responses identified amoeba I with greater surface area; hence, drafted irrelevant description. These responses described the general features of amoeba such as unicellular, irregular in shape, presence of nucleus and vacuoles. However, some of these responses correctly identified the form of amoeba but failed to give the relevant description to support their identification.

Example 1:


I diagram have a greater surface area, because of its shape- omd have flat shape, Pseudopodes are arranged as side to side as compare to the second one-

Example 2:


Ans: Amaebae (I) have greater surface area, have distinct nucleus. Amaeba (I) have proper shape. and (II) have not proper shape. (I) amaeba have ~~not~~ rigid structure.

Question 5:

Give ONE advantage of each of the given features of the respiratory system of human beings.

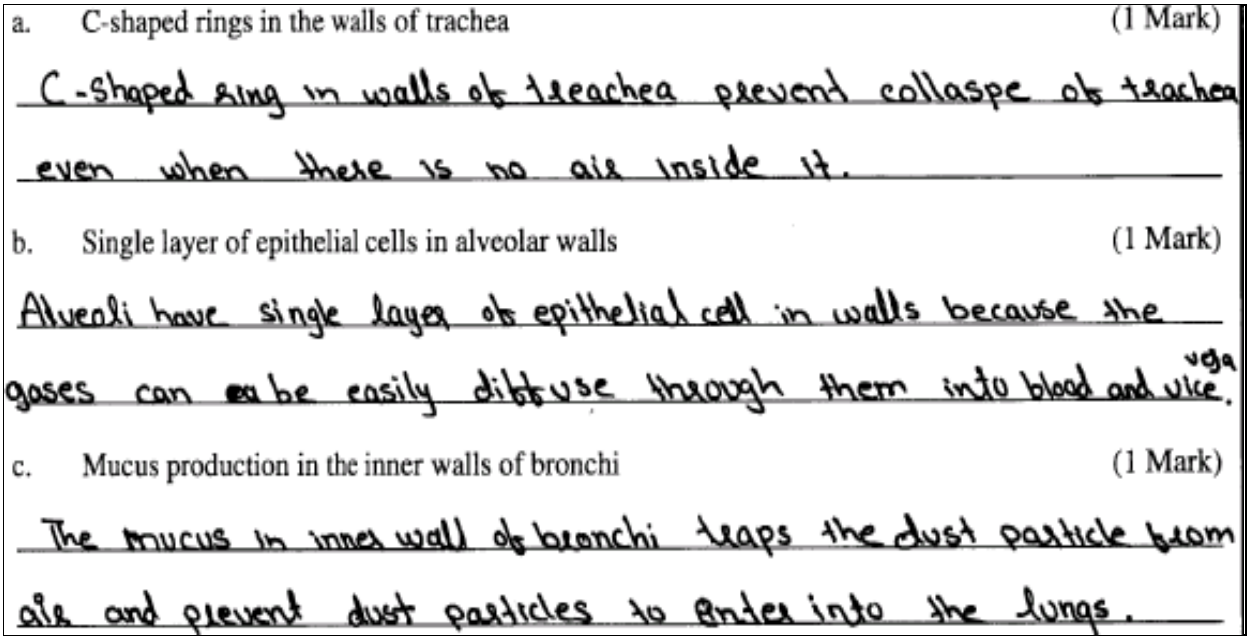
a. C-shaped rings in the walls of trachea

b. Single layer of epithelial cells in alveolar walls

c. Mucus production in the inner walls of bronchi

Better responses exhibited clear understanding of the anatomy and physiology of different organs of human respiratory system and the significance of relationship between structure and function of these organs. Such responses mentioned that C-shaped rings keep the trachea open/ prevent it from collapsing/ provide space for esophagus. Similarly, some of the responses mentioned the benefit of single layer of epithelial cells in alveolar walls is to provides easy access to the diffusion of gases/ provide short diffusion distance. Likewise, the advantage of mucus production in the inner walls of bronchi is to moisten the air/ trap dust particles that escaped from nasal cavity.

Example:


a. C-shaped rings in the walls of trachea (1 Mark)

C-shaped ring in walls of treachea prevent collaspe of trachea even when there is no air inside it.

b. Single layer of epithelial cells in alveolar walls (1 Mark)

Alveoli have single layer of epithelial cell in walls because the gases can ea be easily diffuse through them into blood and vice vga.

c. Mucus production in the inner walls of bronchi (1 Mark)

The mucus in inner wall of bronchi traps the dust particle from air and prevent dust particles to enter into the lungs.

Weaker responses showed misunderstanding of the question, thus, could not fulfill its demand. Such responses mentioned the function of trachea, alveoli and bronchi. For example, the advantage of trachea is to carry gases to the lungs/ alveolus is to exchange gases with blood. Few of these responses mentioned generalised answers such as proper C-shape of tracheal wall, single layer epithelium of alveoli and mucus in the wall of bronchi help in proper functioning in respiration.

Example:

a. C-shaped rings in the walls of trachea (1 Mark)

C-Shaped ring help to pass food from treachea. easily.

b. Single layer of epithelial cells in alveolar walls (1 Mark)

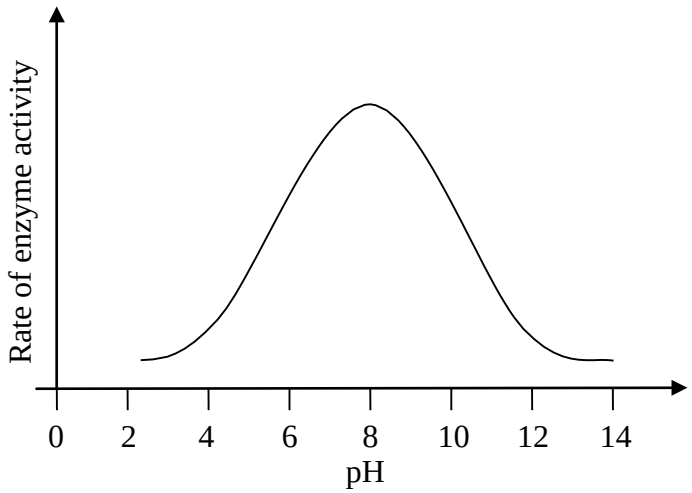
Protect the Inner Parts of alveolar or provide Strength.

c. Mucus production in the inner walls of bronchi (1 Mark)

It provide Smoth-ness that food Pass very easily. to our Stomach. or easily Pass through bronchi.

Question 6:

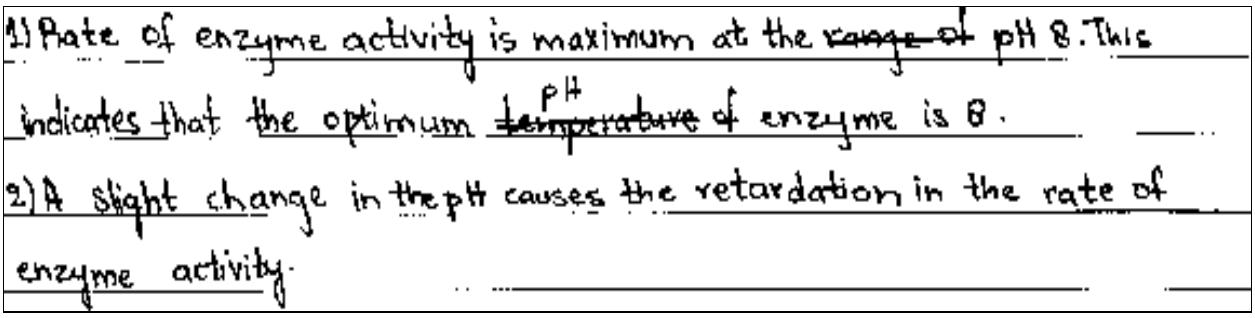
The given graph shows the activity of an enzyme in the human body at different pH.



Extract any TWO interpretations from the given graph.

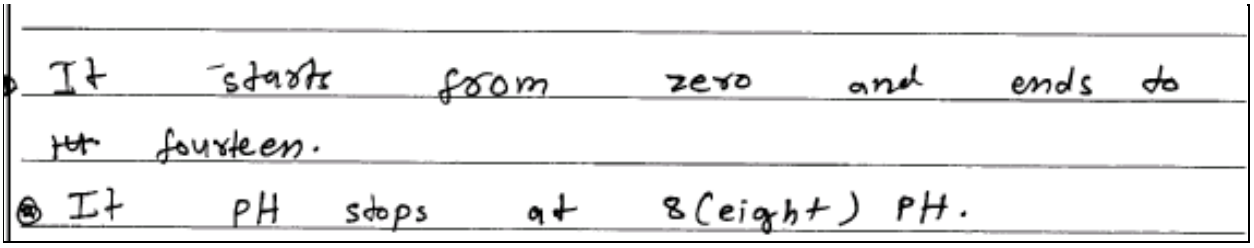
Better responses showed comprehension of graphical representation of the factor affecting the enzyme's activity. Moreover, such responses correctly followed the command word and key word that are 'extract 'and 'interpretation' respectively. These responses extracted points such as rate of the enzyme activity increases with the increase in pH/ enzyme activity is optimum at 8 pH/ the rate of enzyme activity decreases beyond the optimum pH.

Example:



Weaker responses failed to comprehend the command word and key word of the question thus mentioned all the memorised points related to enzymes. Such as enzymes are specific in their action/ they increase rate of reactions/ protein in nature. Few responses mentioned about the pH at which trypsin activity is optimum.

Example:



Better responses displayed a thorough understanding of the structural adaptations of different types of blood vessels. Such responses mentioned that veins have valves to prevent backflow of blood, capillaries have one-cell thick epithelium to ensure shorter distance between capillary and surrounding/ facilitates easy diffusion of materials between blood vessel and tissue and arteries have thick muscular walls as compared to veins to bear blood pressure.

Question 7:

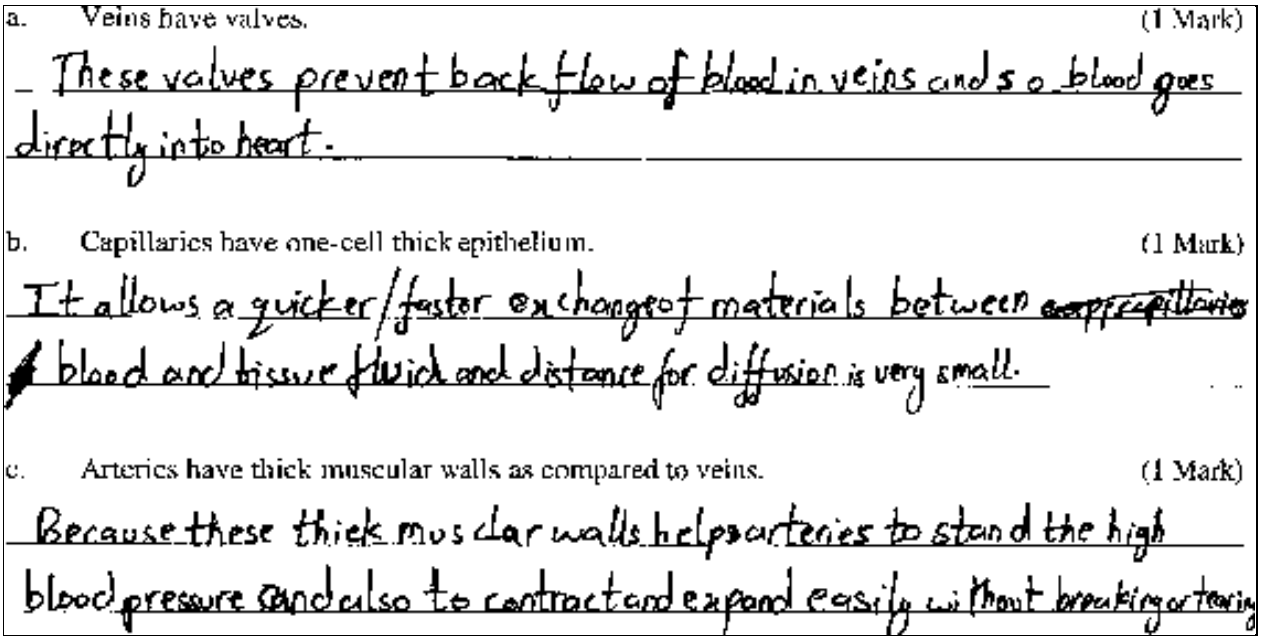
Give reasons for the following adaptations of blood vessels.

a. Veins have valves.

b. Capillaries have one-cell thick epithelium.

c. Arteries have thick muscular walls as compared to veins.

Example:


a. Veins have valves. (1 Mark)

These valves prevent back flow of blood in veins and so blood goes directly into heart.

b. Capillaries have one-cell thick epithelium. (1 Mark)

It allows a quicker/faster exchange of materials between ~~cap~~ capillaries & blood and tissue fluid and distance for diffusion is very small.

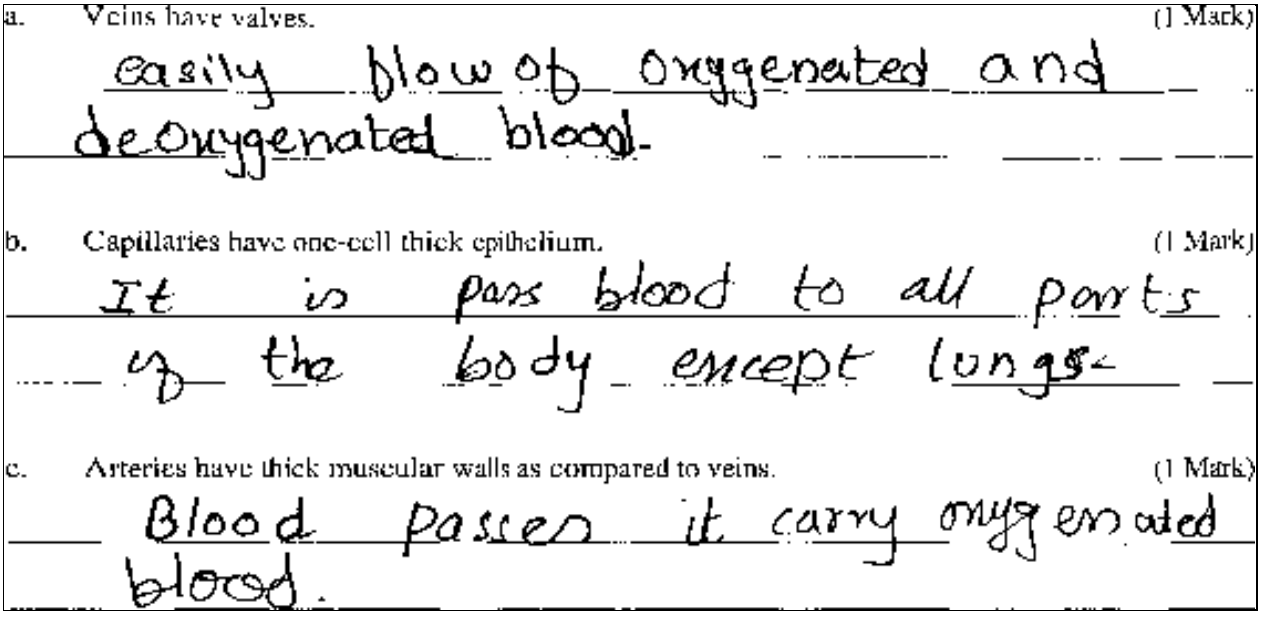
c. Arteries have thick muscular walls as compared to veins. (1 Mark)

Because these thick muscular walls helps arteries to stand the high blood pressure and also to contract and expand easily without breaking or tearing

Weaker responses were unable to comprehend the demand of the question. Such responses provided the function of veins, capillaries and arteries. Moreover, few candidates described the structural features of veins, capillaries and arteries. For example, veins have large lumen, thin layer of smooth muscles, capillaries are composed of single layer and arteries have narrow lumen and thick smooth muscles. Such responses indicate candidates' rote memorisation of the concept of blood vessels.

Teachers are highly recommended to emphasise on relating the structure with function of blood vessels and highlighting the significance of the relationship.

Example 1:


a. Veins have valves. (1 Mark)

easily flow of oxygenated and deoxygenated blood.

b. Capillaries have one-cell thick epithelium. (1 Mark)

It is pass blood to all parts of the body except lungs

c. Arteries have thick muscular walls as compared to veins. (1 Mark)

Blood passes it carry oxygenated blood.

Example 2:

Veins have valves. (1 Mark)

Because veins have ability to carry blood toward heart

Capillaries have one cell thick epithelium. (1 Mark)

Because capillaries are very small in size

Arteries have thick muscular walls as compared to veins. (1 Mark)

Because arteries carry blood away from the heart

Extended Response Questions (ERQs)

The following questions (8 and 9) offered a choice between part **a** and **b**.

Larger number of candidates attempted part 'b' of question 8 as compared to part 'a'. This shows their interest and strong understanding of the concept of 'photosynthesis'. Similarly, in question 9, more preference of candidates was observed in part 'b' than part 'a'. This indicates that candidates are more confident in plant physiology.

Question 8a:

Bacterial cell is classified as a prokaryotic cell while a plant cell as a eukaryotic cell.

With reference to the given classification, mention any THREE similarities and FOUR differences between a bacterial and a plant cell.

Better responses showed a sound understanding of comparison between a bacterial and plant cell. Such responses mentioned the basic similarities including presence of genetic material, i.e. presence of DNA and RNA, cell membrane, ribosomes for protein formation, cell wall and photosynthetic pigment. Likewise, such responses mentioned the important differentiating points such as bacterial cells are smaller in size as compared to plant cells/ the genetic material of bacteria is scattered in the cytoplasm while in plants, it is surrounded by nuclear membrane/ ribosomes in plants are larger as compared to bacteria/ cell wall of bacteria is made up of peptidoglycan while that of plants is made up of cellulose/ bacteria contain mesosomes for ATP synthesis while plants contain mitochondria and plants contain membrane bounded organelles while bacteria lack membrane bounded organelles.

Example:

Bacterial cells are classified in prokaryotic cell while the ~~eukaryotic~~ cell Plant are considered under eukaryotic cells. These both have many Similliarities and many differences; There are also [differences]

* Plant cell	* Bacterial cell
① Plant cells nucleus has a nuclear envelope its chromosomes and DNA are present in their nucleases.	① Bacterial cell donot have a nuclear envelope and their chromosomes and DNA is freely floating in the cytoplasm.
② Plant cell respectively has membrane bound organells; Enample : Mitochondria e.t.c	② Bacterial cell has ~~a~~ also absense of membrane bound organelles. for example : Mitochondria e.t.c
③ Plant cell carries out aerobic respiration in Mitochondria.	③ Bacterial cell carry out anaerobic respiration in its cytoplasm.
④ Plant cell has a cell wall which contain "Cellulose".	④ Bacterial cell also has a cell wall but it contains "peptidoglycan".
⑤ Plant cell has larger cell than Bacterial cell.	⑤ Bacterial cell has smaller cells than Plant cell.

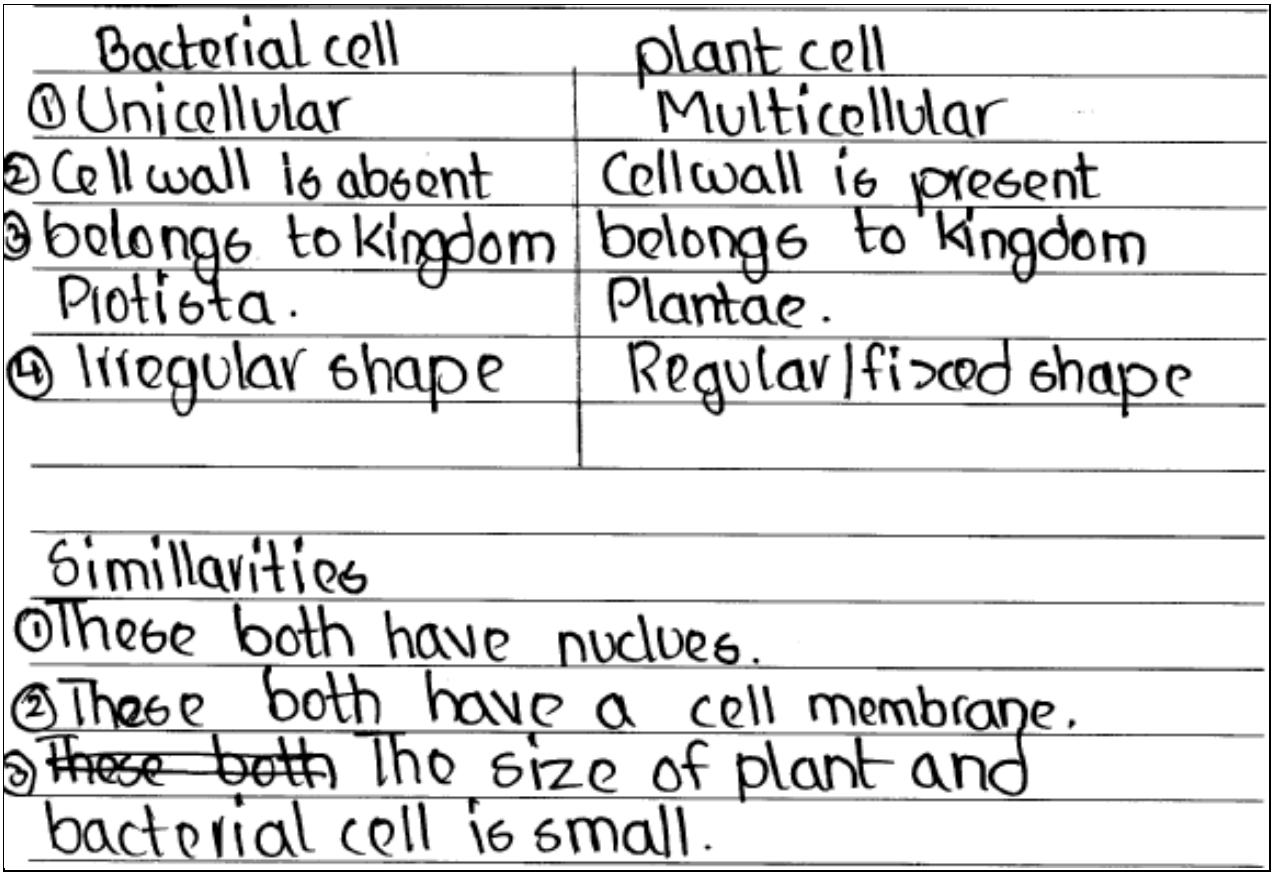
ii) Plant and Bacterial cell also have some simillarities :

Plant Cell	Bacterial cell.
① Plant Cell have Ribosomes present in it that work for protein sythesis.	① Bacterial cell are also having Ribosomes that also work for protein synthesis.
② Plant cell has DNA that is present in nucleus with Chromosomes.	② Bacterial cells are also having DNA that is freely floating with Ribosomes.
③ Plant cells also ~~coo~~ have cell wall that is around outer to cell membrane which protect and provide it strength.	③ Bacterial cells also have Cell Wall that is outer to cell membrane and ~~work~~ serve function to protect inner protoplasm.

Weaker responses overlooked the given stem, therefore, compared the kingdom Prokaryotae (Monera) and Plantae in general. Such responses compared the cellularity, physiology and morphology of plants with the prokaryotes. These responses provided answers like bacteria are unicellular and plants are multicellular/ plants have roots, stem and leaves whereas, bacterial cell do not possesses such structures/ vascular tissues are present in plants and bacteria are devoid of them/ plants are green in colour whereas, bacteria are colourless. Moreover, these responses also mentioned the general characteristics of living organisms in the similarity such as both are organisms and both respire/ grow/ reproduce/ take nutrition and excrete waste materials. Candidates are advised to be conscious about the stem and command word used in the question while drafting their answers.

However, a few responses were able to gauge the demand of question but the candidates failed to draft the correct answer. Such responses provided comparison on the basis of general characteristics of a cell. For example, these responses mentioned that they have cytoplasm, both have organelles and their organelles carry out function.

Example:



Question 8b:

i. Write a balanced chemical equation for photosynthesis.

ii. Explain the role of the THREE limiting factors in photosynthesis.

Better responses stated a balanced chemical equation for photosynthesis including all the reactants and products with their correct number of atoms.

In the second part of the question, candidates clearly stated the limiting factors and provided a thorough explanation of each factor. Moreover, few responses explained showed graphical representation. Most of candidates explained light, carbon dioxide gas and temperature as limiting factors. Such responses explained the effect of limiting factors at different intensities such as with the increase in light, carbon dioxide gas and temperature speed up the rate of photosynthesis.

Example:

b. Photosynthesis:

$$6CO_2 + 12H_2O \longrightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Limiting factor are the factors the deficiency of which can affect the rate of a particular reaction.

i) Intensity of light:

The rate of photosynthesis increase with the increase in intensity of light and decrease with the decrease in the intensity of light. But at much higher light intensity rate of photosynthesis becomes constant.

ii) Temperature:

Rate of photosynthesis increases with the increase in temperature, and decrease with the decrease in temperature But if the light intensity is low increasing the temperature has little influence on photosynthesis.

iii) Concentration of carbondioxide:

Rate of photosynthesis increase with the increase in concentration of CO_2 and vice versa. Increase of CO_2 concentration abar a cetain level causes the closure of stomata and decreases the rate of photosynthesis.

Role of these limiting factors:

i) Light is required for light reactions to occur and to break water molecules. So ATP and NADH are formed, and without the formation of these dark reactions won't occur and glucose will not be formed.

ii) Carbondioxide is required as it is a raw material for photosynthes food/glucose is formed by reduction of CO_2, so food won't be formed without it.

iii) Temperature; high temperature means more light, and means more photosynthesis, more transpiration so more water for photosynthesis.

Weaker responses provided either completely incorrect or incomplete chemical equation. Furthermore, some responses stated the reactants and products in words. Most of the candidates correctly identified the limiting factors but were unable to provide correct explanation of the identified factors. These candidates explained the factors which affect the rate of transpiration in plant, such as humidity, opening and closing of stomata and surface area of leaves.

Candidates' insufficient knowledge and understanding of the role of temperature, carbon dioxide gas and light was evident. Teachers are highly recommended to integrate the related practical activities given in the syllabus for better understanding of theory.

Example:

b)i. Balanced chemical equation for photosynthesis is in In the form of words:

sunlight + water + carbon dioxide → glucose + oxygen

symbols form

sunlight + H_2O + CO → glucose + O_2

b)ii. Limiting factor: Limiting factor is the factor that controls the process of photosynthesis

*) The role of three limiting factors in photosynthesis.

(1) Humidity: Increase in the rate of Humidity effects the rate of photosynthesis that there will be more humidity so the rate of photosynthesis would be increase more glucose would be forming and if the rate of humidity would be low the rate of photosynthesis would be normal.

*) Greater surface area: There will be greater surface area so the rate of photosynthesis would be because if greater surface area the parts of plant will be direct contact with the environment and cell would be able to exchange the gases easily because have no respiratory system.

3) Properly rate of photosynthesis will done when all the essential product would be their like sunlight, water, carbon dioxide and all the things would be normal and optimum quantity so if some will really effect the rate of photosynthesis so there would be more production of oxygen. And glucose. the glucose will utilizes by the plants and oxygen for us.

Question 9a:

Describe any TWO structural features of each of the following organs of the digestive system of human beings which help them to function efficiently.

i. Stomach

ii. Small intestine

Better responses were able to correctly identify the structural features of stomach and small intestine and described the relationship of identified structure with their function. With reference to stomach majority of the candidates described the muscles and gastric glands. Moreover, other responses highlighted the mucosal layer and sphincter. Such responses articulated the structure and function very clearly such as stomach muscles help in partial digestion of the food/ the churning action of the stomach muscles physically breaks down the food/ gastric glands of stomach wall release acids and enzymes for the chemical breakdown of food/ the enzyme pepsin is responsible for protein breakdown. Pyloric sphincter of stomach releases food into the small intestine in a controlled and regulated manner/ mucous cells secrete the alkaline mucous for shielding the epithelium from hydrochloric acid.

Similarly, most of the candidates described the villi and intestinal glands of small intestine. And few of the candidates also described the microvilli, circular folds, smooth muscles and long length. These responses described the function such as villi in the small intestine increase surface area for the maximum absorption. Intestinal glands secrete intestinal juice that contains many enzymes for the complete digestion of carbohydrates, proteins and fats/ circular folds are large flaps that project into the lumen of the small intestine/ the circular folds slow the passage of food along the intestines/ the epithelial cells lining the villi are covered with minute projections called microvilli which increase surface area for absorption.

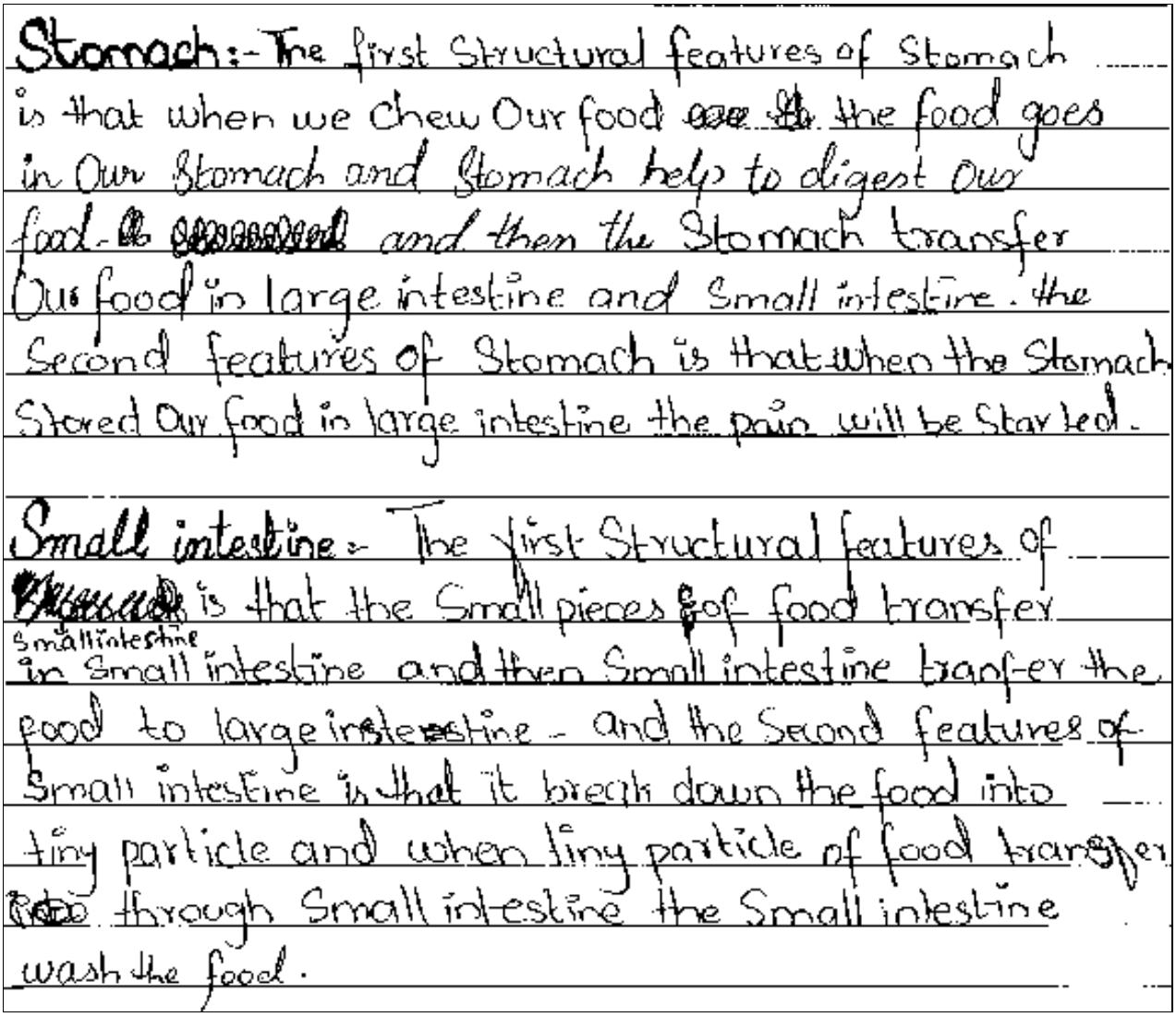
Example:

(a) The stomach is ~~present~~ on the left side of abdomen underneath diaphram. Stomach is divided into two portions: **Cardiac portion** & **Pyloric portion**. The cardiac portion would produce **gastric juice** that consists of **HCL**, **mucous** & **peptigen** **peptinogen**. The HCl would kill any bacteria or harmful thing and also activates the **peptinogen** into its active form (pepsin) that would digest digest **protein** in food into ~~peptide~~ **polypeptides**. The walls of stomach is also protein but as do not digest as ~~the mucous~~ the mucous covers up the wall of stomach the pepsin doesn't act on it. the ~~f~~ partially breakdown of protein happens in Pyloric portion. Then stomach walls also contains **Smooth muscles** that helps in **churning** Contraction & mixing of food ~~it~~ take place & a soup like thing is made (**Chyme**). Then the food goes to the duodenum where it is further breakdown and Bile from liver helps in emulsification of Lipids. & pancreatic juice containing lipase, trypsin, amalayse digests the carbohydrates, protein & Lipids intenstial juice further breakes down ~~lipid~~ lipids into fatty acids & glycerol then it went towards jejenium more digestion take place than in ileium absoptation of digested food take place here it has inner folds in which finger like projections **villi** is present. That increases the surface area & these villi have lots of cappilaries on them & vessel of lymphatic system, **lacteal** is present. The ~~food~~ amino acids & simple sugars are absorbed in blood cappilaries & carried toward liver through hepatic portal vein for some filteration & extra food is stored after that ~~that~~ blood is taken towards heart via hepatic vein & the fatty acid & glycerol in ~~stomach~~ small intenstine is taken towards lymphatic duct where it could defuse in blood stream. It consists of important veins that would help it to transport digested food & absorption of food material into blood.

Weaker responses were unable to grasp the demand of the question. In the majority of responses a general trend was evident such that candidates described the function of stomach and small intestine. Such responses merely highlighted the structural features of stomach and small intestine. Hence, most of these responses described the process of digestion from oral cavity to the large intestine.

Teachers are highly recommended to inculcate the concept of structure-function relationship while teaching human organ systems.

Example:


Stomach:- The first Structural features of Stomach is that when we chew Our food the food goes in Our Stomach and Stomach help to digest Our food and then the Stomach transfer Our food in large intestine and Small intestine. the Second features of Stomach is that when the Stomach Stored Our food in large intestine the pain will be Started.

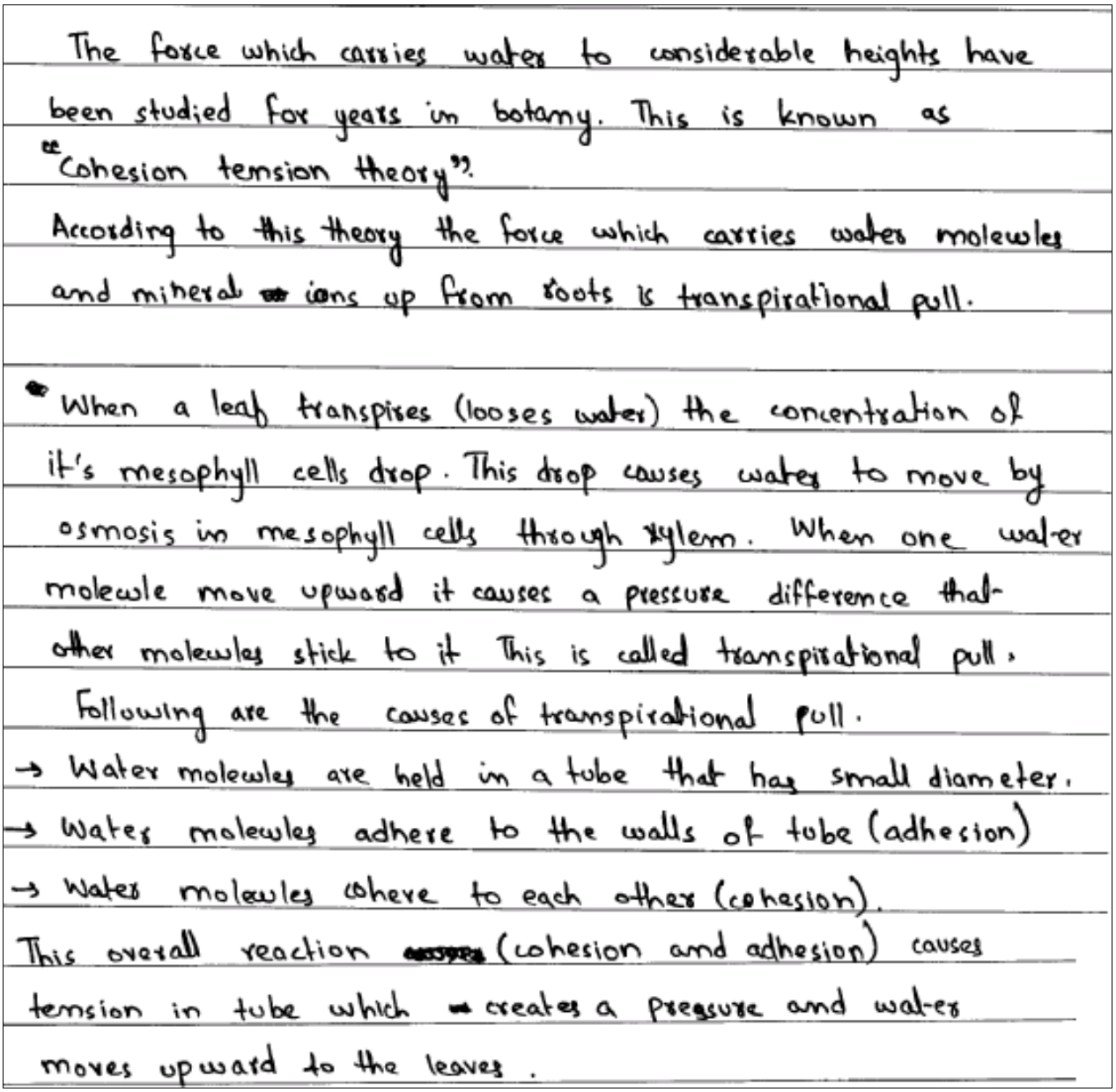
Small intestine:- The first Structural features of is that the Small pieces of food transfer in Small intestine (Small intestine) and then Small intestine transfer the food to large intestine - and the Second features of Small intestine is that it break down the food into tiny particle and when tiny particle of food transfer through Small intestine the Small intestine wash the food.

Question 9b:

In plants, water continuously moves from roots to xylem and transpires through leaves. Explain the transportation of water in plants with reference to transpiration pull.

Better responses provided a thorough explanation of transpiration pull. Moreover, steps of transpiration pull were explained in a coherent way. Such responses explained the development of pressure difference by transpiration, role of osmosis in the movement of water from xylem to mesophyll cells and creation of pulling force. Moreover, these steps were supported by the relevance of structure of xylem and cohesion adhesion property of water.

Example:



The force which carries water to considerable heights have been studied for years in botany. This is known as "Cohesion tension theory".

According to this theory the force which carries water molecules and mineral ions up from roots is transpirational pull.

When a leaf transpires (looses water) the concentration of it's mesophyll cells drop. This drop causes water to move by osmosis in mesophyll cells through xylem. When one water molecule move upward it causes a pressure difference that other molecules stick to it This is called transpirational pull.

Following are the causes of transpirational pull.

→ Water molecules are held in a tube that has small diameter.

→ Water molecules adhere to the walls of tube (adhesion)

→ Water molecules cohere to each other (cohesion).

This overall reaction (cohesion and adhesion) causes tension in tube which creates a pressure and water moves upward to the leaves.

Weaker responses mostly focused on the stem of the question. Such responses elaborated the stem in different ways. In this case, candidates drafted the definition of transpiration, restated the stem and provided the explanation of factors affecting rate of transpiration. Moreover, some of these responses described the disadvantages of excessive transpiration rate for plants while few responses related transpiration with photosynthesis.

However, a few of candidates were able to provide a brief description of the process of transpiration but were not able to provide a stepwise explanation; hence, such responses were not awarded full marks.

Teachers are recommended to use flow charts and concept maps for the better understanding and stepwise memorization of the concept.

Example:

Transpiration pull is a suction force through which the water from roots is drawn upward to Arial parts of plant and evaporate. More transpiration can cause a plant to wilt. Transpiration gives cooling effect to the plant and prevents the plant from heating up due to hot sun. It helps in Photosynthesis. Excessive transpiration can result in wilting of plant. Wilting is a state resulted from excesive loss of water by the plant. Wilting has Advantage and disadvantages. The advantage is that when a plant loses water it guard cell become flaccid and Stoma closes and the Surface area of plant decreases (this reduces the surface area facing sun) this prevents the plant from a ~~pre~~ permanent damage. Disadvantage of wilting includes that the rate of photosynthesis decreases as because CO_2 cannot ~~ene~~ enter the guard cell because guard cells are closed so no more glucose can be produces and due to water as a limiting factor the photosynthesis rate decreases.